

Video Delivery, Protection & Player Experience

Context

Neuroflix makes corporate training suck less. We use agentic AI pipelines to generate engaging video-based training content designed to maximize knowledge retention. Organizations create training courses on Neuroflix and enroll their employees to complete them.

Everything in Neuroflix revolves around the video player. Each lesson is a video, and during playback, checkpoint questions pop up at specific timestamps that learners must answer before continuing. The player is the product. If the video experience is slow, clunky, or feels like a 2010 corporate training tool, nothing else matters.

We need to solve three problems:

1. **The player must feel like Netflix or YouTube.** Fast buffering. Thumbnail previews when scrubbing the timeline. A responsive, intuitive experience on both desktop and mobile. Smooth seeking with visual feedback for what's loaded and what's playing.
 2. **Videos must be protected.** Enterprise clients pay for this content. Learners should not be able to trivially download the videos. Opening the browser developer tools and grabbing a URL should not give someone a downloadable copy of the video.
 3. **Videos must be served fast and globally.** Video should be delivered through a CDN, not proxied through application servers. Buffering should be near-instant. Playback should adapt to the viewer's connection speed.
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1. Player Experience

The video player needs to feel modern and polished. It should be a custom player (not native browser controls) that works excellently on both desktop and mobile.

Timeline and Seeking

- When hovering over the progress bar (or long-pressing on mobile), show a **thumbnail preview** of the video frame at that position.
- Display the **buffered range** visually on the progress bar, so the viewer can see how far ahead the video has loaded.
- Show **checkpoint markers** on the timeline where questions will appear during playback.

- Seeking should be smooth, with immediate visual feedback.

Checkpoint Questions

- At configured timestamps, the video pauses and a multiple-choice question overlay appears.
- Each question has four answer options. The learner selects one answer before playback continues.
- This is a core feature of the Neuroflix player and must be preserved in whatever player architecture is chosen.

Watermarking

- A visible, semi-transparent watermark overlay should be displayed during playback, showing identifying information about the viewer (e.g., their email and organization).
- The watermark deters screen recording by embedding the viewer's identity in any recording.
- The watermark position should shift periodically to make it harder to crop out.

Mobile

- The player must be fully responsive and work well on mobile devices in both portrait and landscape.
- Touch targets must be large enough for comfortable use (minimum 44px per Apple Human Interface Guidelines).
- Controls should auto-hide during playback and reappear on tap.
- Buffering and loading states should feel native and fast.

2. Streaming & Download Prevention

Serving a single MP4 file means the browser receives one URL that can be copied and used to download the entire video. This section is about making casual downloading meaningfully harder.

The goal is to move from direct MP4 delivery to a segmented streaming approach where the browser never has a single URL that represents the full video. The Network tab should show many short-lived segment requests, not one downloadable file.

This also unlocks adaptive bitrate streaming, where the player automatically adjusts video quality based on the viewer's connection speed. This is critical for mobile users on variable connections and is a core part of why Netflix and YouTube feel smooth.

Thumbnail sprite sheets for the timeline previews in Part 1 are typically generated during the transcoding step, so the streaming pipeline and player experience should be planned together.

3. CDN Infrastructure

Videos need to be stored in private cloud storage and served through a CDN with access control. Only authenticated, authorized users should receive a valid URL, and that URL should expire after a short window. Video bytes should never flow through the application server. The server handles auth, generates a short-lived signed URL, and redirects the client directly to the CDN.

This layer should also handle the upload pipeline: when a video is added to the platform, it goes into the CDN's backing storage, and the stored reference is used for all subsequent serving. At scale, video bandwidth is the dominant cost line, so the choice of storage and CDN provider matters. Note that egress fees vary dramatically between providers (some charge per-GB, others have zero egress).

Deliverables

Build a working video player application that demonstrates all three areas above. Use your own test video(s). The three parts are sequential in importance: the player experience is the most visible, streaming builds protection and performance on top of it, and CDN infrastructure ties the production architecture together.

Deadline: Friday, May 29th at 8 PM IST.